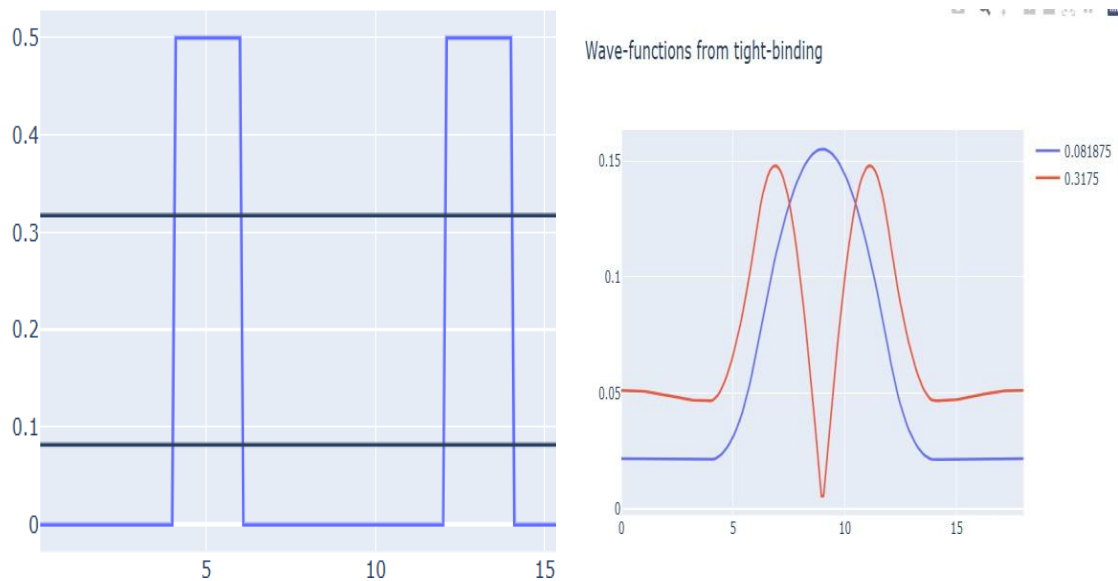


EXP 5: DEMONSTRATION OF QUANTUM TUNNELING

Aim:

To simulate the band structure of nanowire for various materials.

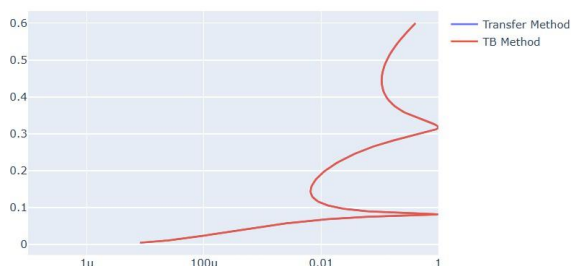
Potential Energy vs Distance



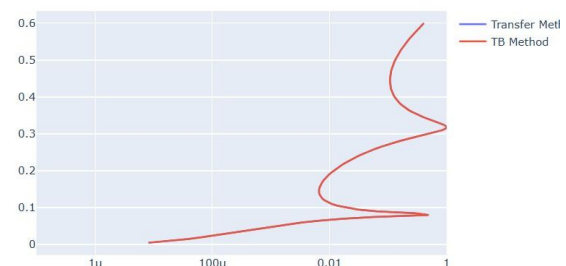
This graph illustrates the variation of potential energy with distance in a periodic potential structure. The alternating high and low potential regions represent the barriers and wells experienced by electrons as they travel through the material. The horizontal energy levels indicate the possible energies of charge carriers within the structure. This potential profile is widely used to study quantum confinement, carrier motion, and energy band formation in semiconductor nanostructures.

Case study 1 : Analysis the transmission co-efficient Vs Energy characteristics on refined grid

Transmission Coefficient Vs Energy on refined grid



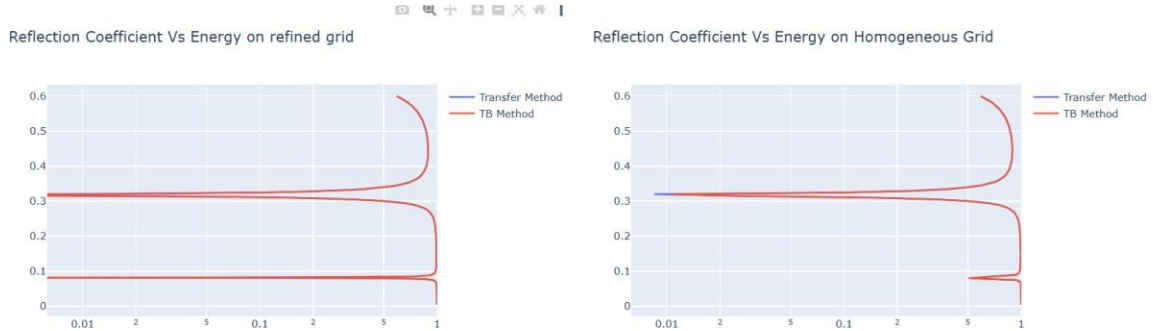
Transmission Coefficient Vs Energy on Homogeneous Grid



This graph presents the transmission coefficient of electrons at different energy levels using both the Transfer Matrix and Tight-Binding (TB) methods. The close overlap between the two

curves confirms that both numerical techniques produce similar predictions for electron transmission. The increase and decrease in transmission indicate the energy ranges where electrons can easily pass through the potential barriers. This analysis is important for evaluating quantum transport in nanoscale semiconductor devices.

Case study 2 : Analysis the Reflection co-efficient Vs Energy on Homogeneous Grid



This graph shows the reflection coefficient as a function of electron energy for both the Transfer Matrix and TB methods. The results indicate how much of the incoming electron wave is reflected by the periodic potential barriers at different energies. Similar trends in both methods demonstrate the consistency and reliability of the simulation. This study helps in understanding electron scattering and optimizing the design of quantum and semiconductor devices.

This comparison evaluates the simulation results obtained using refined and homogeneous computational grids. Both grid types produce nearly identical transmission and reflection characteristics, indicating that the numerical solution is stable and accurate. The refined grid provides higher precision near regions of rapid variation, while the homogeneous grid offers faster computation. Such comparisons are useful for selecting the most suitable numerical approach for quantum transport simulations.

CONCLUSION: The band structure of nanowires for different materials was successfully simulated using the nanoHUB ABACUS tool. The simulation showed that reducing the wire dimensions to the nanoscale causes quantum confinement, resulting in discrete energy levels and changes in the electronic band structure. The band structures of InAs and AlGaAs circular nanowires were studied and compared. Thus, the experiment helps in understanding the quantum behavior and electronic properties of nanoscale materials, which are important for designing nanowire-based electronic and nanoelectronic devices.